```r
# ------------------------------------------------------------------------
# 0. SETUP
# ------------------------------------------------------------------------

needed_packages <- c("ggplot2", "dplyr", "broom")

for (pkg in needed_packages) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg)
  }
}

library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.6.1
```

```r
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.6.1


Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(broom)
```

```
Warning: package 'broom' was built under R version 4.6.1
```

```r
options(digits = 4)
```

```r
# ========================================================================
# 1. Understanding the Data
# ========================================================================
```

```r
# The original public source is:
# https://www4.stat.ncsu.edu/~boos/var.select/diabetes.tab.txt
data_file <- "diabetes.tab.txt"

if (!file.exists(data_file)) {
  message("diabetes.tab.txt was not found in the current working directory.")
  message("Please choose the supplied diabetes.tab.txt file.")
  data_file <- file.choose()
}


diabetes <- read.table(data_file, header = TRUE, sep = "\t")

# Convert column names to lower case so they are easier to type consistently
names(diabetes) <- tolower(names(diabetes))

# The source dataset uses 'y' as the disease-progression outcome.
if ("y" %in% names(diabetes)) {
  diabetes <- diabetes %>% rename(progression = y)
}

# first few rows.
head(diabetes)
```

```
  age sex  bmi  bp  s1    s2 s3 s4    s5 s6 progression
1  59   2 32.1 101 157  93.2 38  4 4.860 87         151
2  48   1 21.6  87 183 103.2 70  3 3.892 69          75
4  24   1 25.3  84 198 131.4 40  5 4.890 89         206
5  50   1 23.0 101 192 125.4 52  4 4.290 80         135
6  23   1 22.6  89 139  64.8 61  2 4.190 68          97
```

```r
# Number of rows and columns.
dim(diabetes)
```

```
[1] 442  11
```

```r
# Variable names.
names(diabetes)
```

```
 [1] "age"         "sex"         "bmi"         "bp"          "s1"
 [6] "s2"          "s3"          "s4"          "s5"          "s6"
```

```
[11] "progression"
```

```
# Structure and data types.
str(diabetes)
```

```
'data.frame':   442 obs. of  11 variables:
 $ age        : num  59 48 72 24 50 23 36 66 60 29 ...
 $ sex        : num  2 1 2 1 1 1 2 2 2 1 ...
 $ bmi        : num  32.1 21.6 30.5 25.3 23 22.6 22 26.2 32.1 30 ...
 $ bp         : num  101 87 93 84 101 89 90 114 83 85 ...
 $ s1         : num  157 183 156 198 192 139 160 255 179 180 ...
 $ s2         : num  93.2 103.2 93.6 131.4 125.4 ...
 $ s3         : num  38 70 41 40 52 61 50 56 42 43 ...
 $ s4         : num  4 3 4 5 4 2 3 4.55 4 4 ...
 $ s5         : num  4.86 3.89 4.67 4.89 4.29 ...
 $ s6         : num  87 69 85 89 80 68 82 92 94 88 ...
 $ progression: num  151 75 141 206 135 97 138 63 110 310 ...
```

```
# The variable 'sex' is numerically coded in the source data, but it represe
# categories rather than a continuous measurement.
unique(diabetes$sex)
```

```
[1] 2 1
```

```
diabetes$sex <- factor(diabetes$sex)

# Count missing values in each variable.
colSums(is.na(diabetes))
```

```
       age        sex        bmi         bp         s1         s2
         0          0          0          0          0          0
        s3         s4         s5         s6 progression
         0          0          0          0          0
```

```
# Total number of missing values.
sum(is.na(diabetes))
```

```
[1] 0
```

```
# Check for duplicated rows.
sum(duplicated(diabetes))
```

```
[1] 0
```

```r
# Basic summary of every variable.
summary(diabetes)
```

```
      age        sex           bmi              bp               s1        
 Min.   :19.0   1:235   Min.   :18.0   Min.   : 62.0   Min.   : 97  
 1st Qu.:38.2   2:207   1st Qu.:23.2   1st Qu.: 84.0   1st Qu.:164  
 Median :50.0           Median :25.7   Median : 93.0   Median :186  
 Mean   :48.5           Mean   :26.4   Mean   : 94.6   Mean   :189  
 3rd Qu.:59.0           3rd Qu.:29.3   3rd Qu.:105.0   3rd Qu.:210  
 Max.   :79.0           Max.   :42.2   Max.   :133.0   Max.   :301  
       s2              s3              s4              s5              s6        
 Min.   : 41.6   Min.   :22.0   Min.   :2.00   Min.   :3.26   Min.   : 58.0  
 1st Qu.: 96.0   1st Qu.:40.2   1st Qu.:3.00   1st Qu.:4.28   1st Qu.: 83.2  
 Median :113.0   Median :48.0   Median :4.00   Median :4.62   Median : 91.0  
 Mean   :115.4   Mean   :49.8   Mean   :4.07   Mean   :4.64   Mean   : 91.3  
 3rd Qu.:134.5   3rd Qu.:57.8   3rd Qu.:5.00   3rd Qu.:5.00   3rd Qu.: 98.0  
 Max.   :242.4   Max.   :99.0   Max.   :9.09   Max.   :6.11   Max.   :124.0  
  progression
 Min.   : 25  
 1st Qu.: 87  
 Median :140  
 Mean   :152  
 3rd Qu.:212  
 Max.   :346  
```

```r
# ============================================================================
# EXPLORE THE DATA
# ============================================================================

describe_numeric <- function(x) {
  c(
    n = sum(!is.na(x)),
    mean = mean(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    sd = sd(x, na.rm = TRUE),
    min = min(x, na.rm = TRUE),
    q1 = quantile(x, 0.25, na.rm = TRUE),
    q3 = quantile(x, 0.75, na.rm = TRUE),
    max = max(x, na.rm = TRUE),
    iqr = IQR(x, na.rm = TRUE)
  )
}
```

```r
# Descriptive statistics for the outcome.
describe_numeric(diabetes$progression)
```

```
     n    mean median     sd    min q1.25% q3.75%    max    iqr
442.00 152.13 140.50  77.09  25.00  87.00 211.50 346.00 124.50
```

```r
# Descriptive statistics for BMI.
describe_numeric(diabetes$bmi)
```

```
      n    mean  median      sd     min  q1.25%  q3.75%     max     iqr
442.000  26.376  25.700   4.418  18.000  23.200  29.275  42.200   6.075
```

```r
# Descriptive statistics for age.
describe_numeric(diabetes$age)
```

```
     n   mean median     sd    min q1.25% q3.75%    max    iqr
442.00  48.52  50.00  13.11  19.00  38.25  59.00  79.00  20.75
```

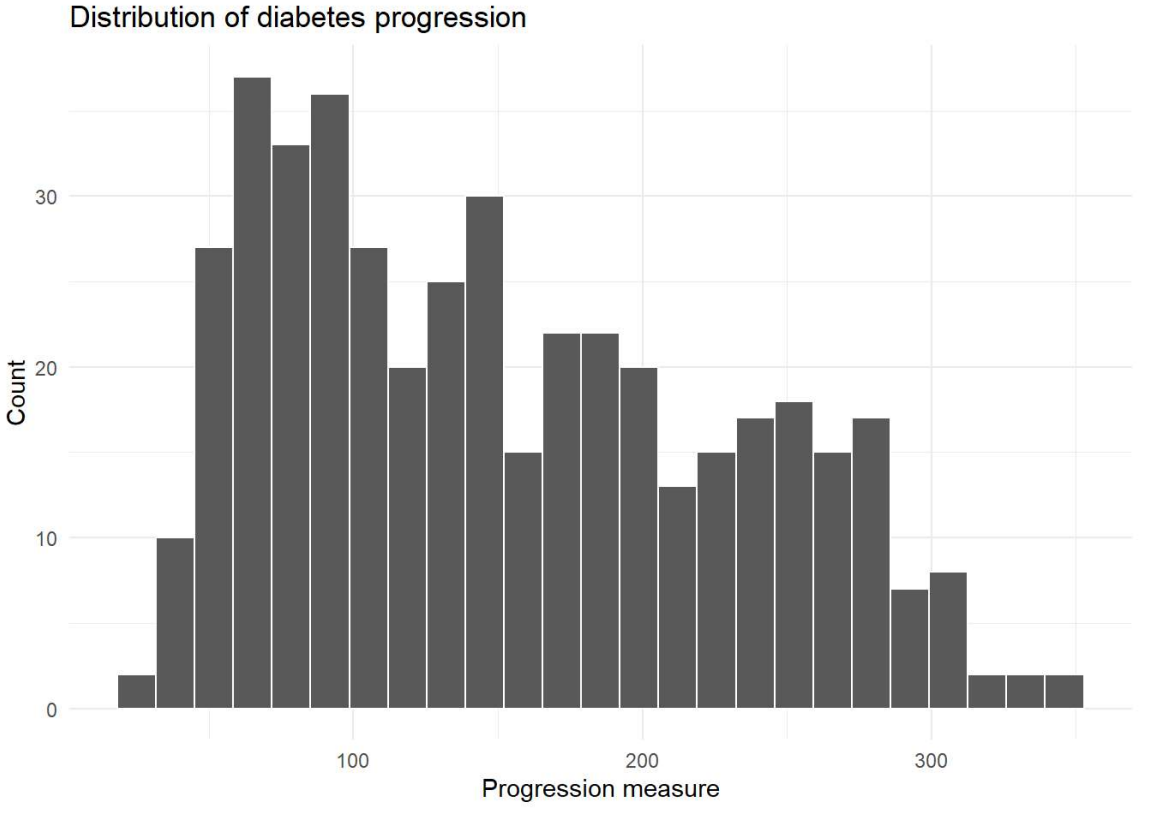
```r
# A compact descriptive table for all numeric variables.
descriptive_table <- diabetes %>%
  summarise(across(
    where(is.numeric),
    list(
      mean = ~mean(.x, na.rm = TRUE),
      median = ~median(.x, na.rm = TRUE),
      sd = ~sd(.x, na.rm = TRUE),
      min = ~min(.x, na.rm = TRUE),
      max = ~max(.x, na.rm = TRUE)
    )
  ))

print(descriptive_table)
```

```
  age_mean age_median age_sd age_min age_max bmi_mean bmi_median bmi_sd
bmi_min
1    48.52         50  13.11      19      79    26.38       25.7  4.418
18
  bmi_max bp_mean bp_median bp_sd bp_min bp_max s1_mean s1_median s1_sd
s1_min
1    42.2   94.65        93 13.83     62    133   189.1       186 34.61
97
  s1_max s2_mean s2_median s2_sd s2_min s2_max s3_mean s3_median s3_sd
s3_min
```

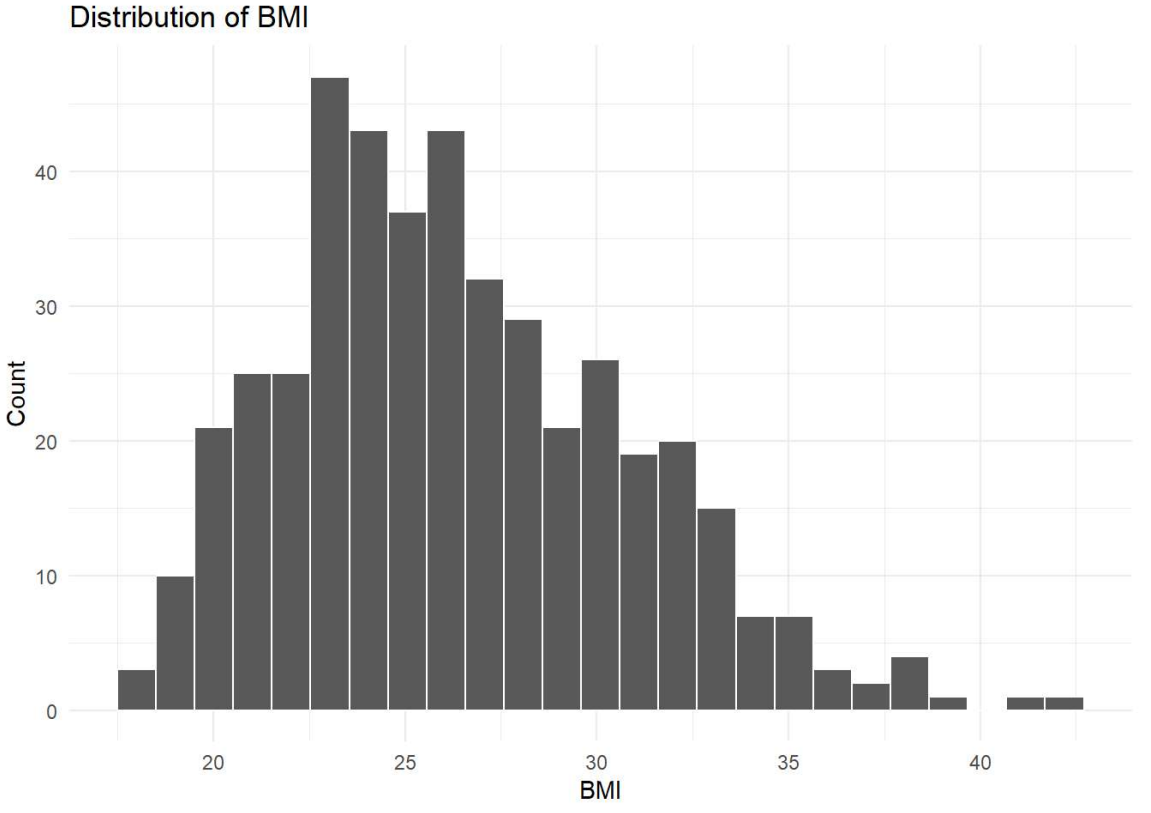
```
1    301   115.4       113 30.41   41.6  242.4   49.79        48 12.93
22
  s3_max s4_mean s4_median s4_sd s4_min s4_max s5_mean s5_median  s5_sd
s5_min
1     99    4.07         4  1.29      2   9.09   4.641      4.62 0.5224
3.258
  s5_max s6_mean s6_median s6_sd s6_min s6_max progression_mean
1  6.107   91.26        91  11.5     58    124            152.1
  progression_median progression_sd progression_min progression_max
1              140.5          77.09              25              346
```

```r
# Distribution of diabetes progression.
ggplot(diabetes, aes(x = progression)) +
  geom_histogram(bins = 25, colour = "white") +
  labs(
    title = "Distribution of diabetes progression",
    x = "Progression measure",
    y = "Count"
  ) +
  theme_minimal()
```

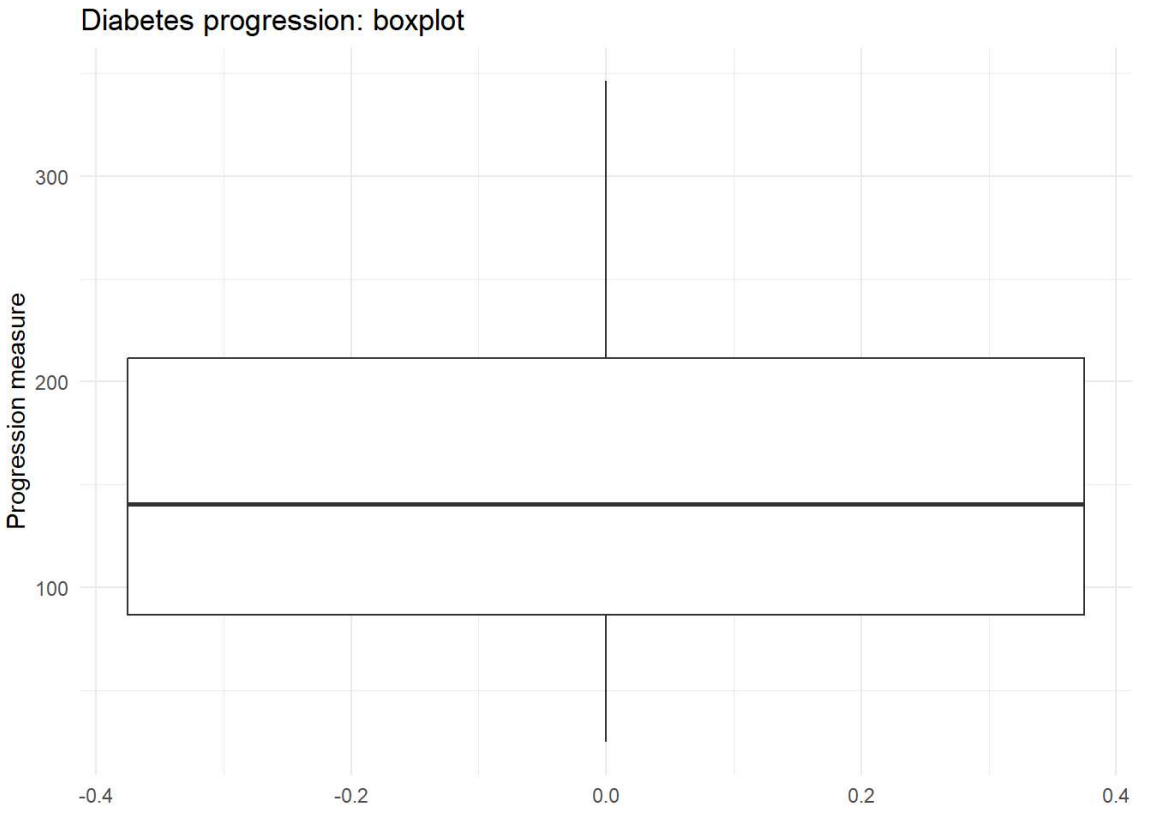



```r
# Distribution of BMI.
ggplot(diabetes, aes(x = bmi)) +
```

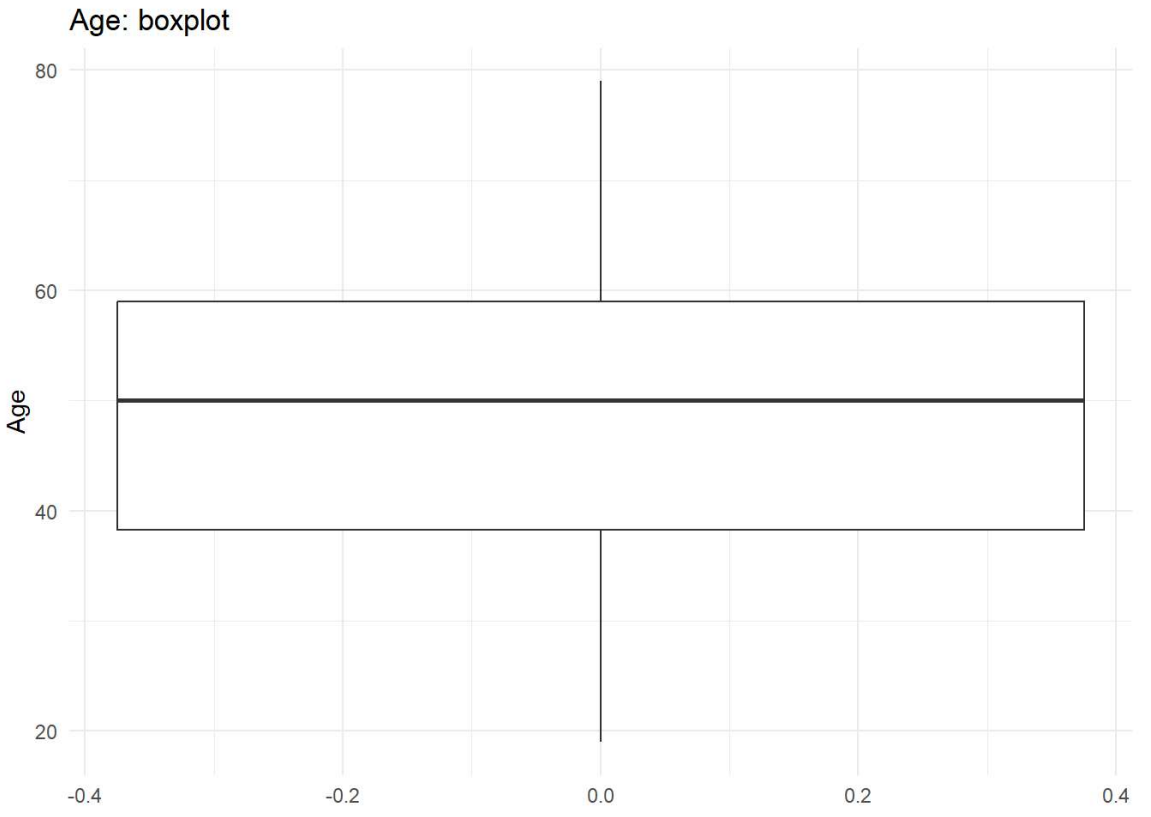
```
  geom_histogram(bins = 25, colour = "white") +
  labs(
    title = "Distribution of BMI",
    x = "BMI",
    y = "Count"
  ) +
  theme_minimal()
```



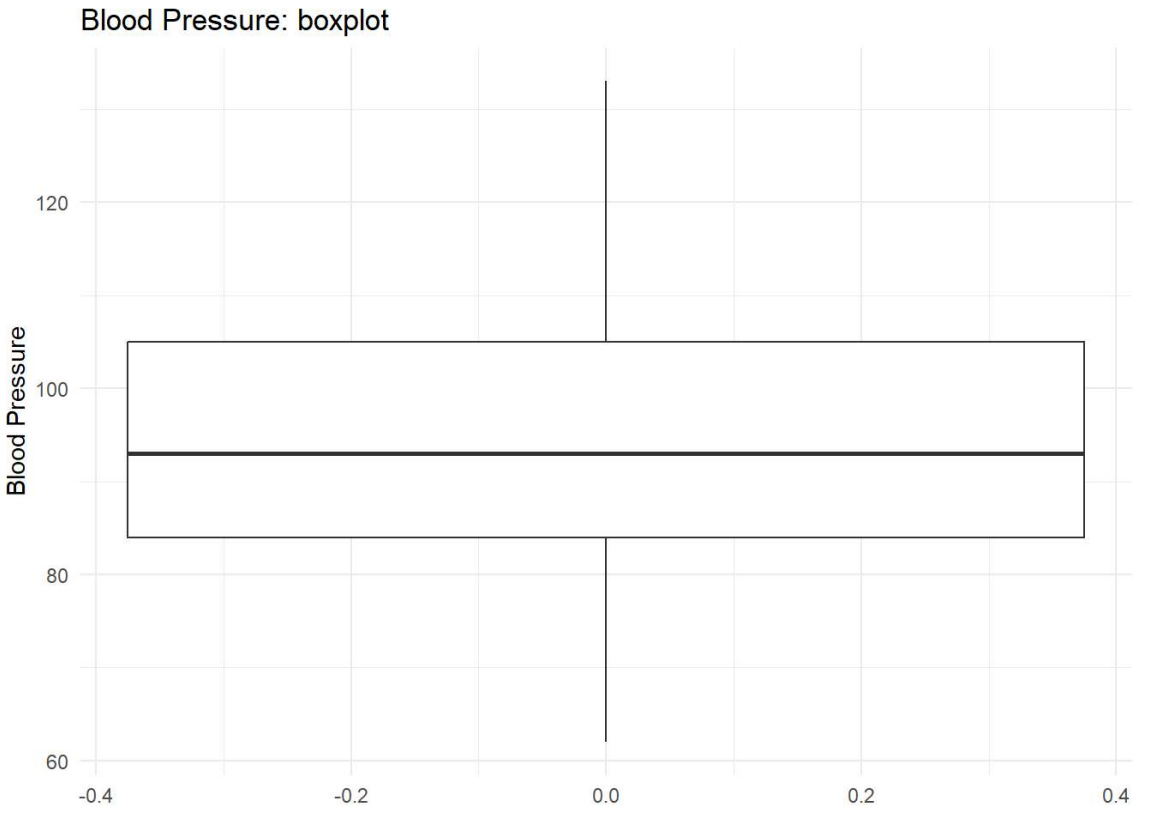

```
# One additional useful visualisation: boxplot of progression.
ggplot(diabetes, aes(y = progression)) +
  geom_boxplot() +
  labs(
    title = "Diabetes progression: boxplot",
    y = "Progression measure",
    x = NULL
  ) +
  theme_minimal()
```




```r
ggplot(diabetes, aes(y = age)) +
  geom_boxplot() +
  labs(
    title = "Age: boxplot",
    y = "Age",
    x = NULL
  ) +
  theme_minimal()
```




```r
ggplot(diabetes, aes(y = bp)) +
  geom_boxplot() +
  labs(
    title = "Blood Pressure: boxplot",
    y = "Blood Pressure",
    x = NULL
  ) +
  theme_minimal()
```




```r
# ============================================================================
#  INVESTIGATE RELATIONSHIPS
# ============================================================================

numeric_data <- diabetes %>% select(where(is.numeric))

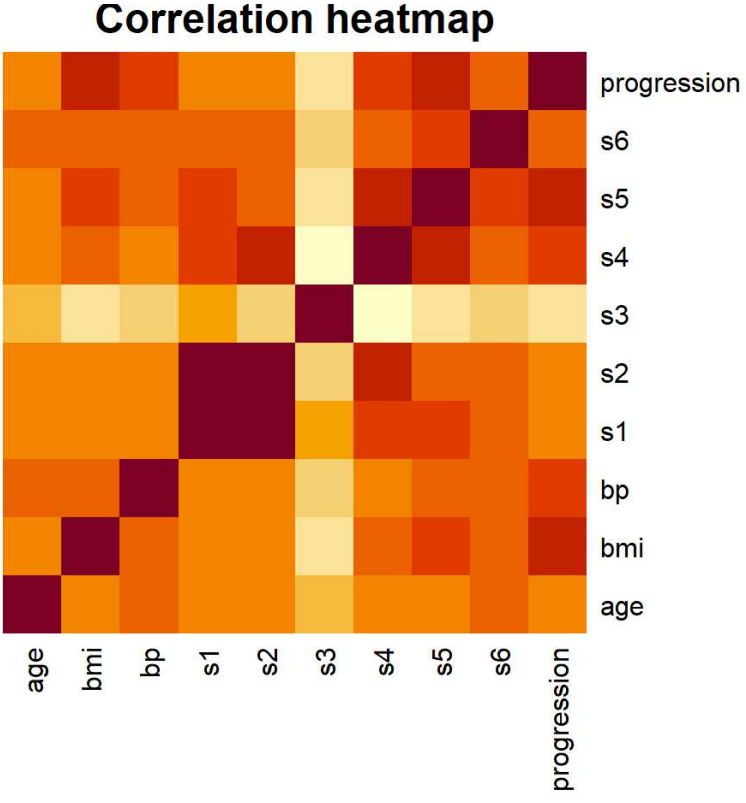
# Correlation matrix.
cor_matrix <- cor(numeric_data, use = "complete.obs", method = "pearson")
round(cor_matrix, 2)
```

	age	bmi	bp	s1	s2	s3	s4	s5	s6	progression
age	1.00	0.19	0.34	0.26	0.22	-0.08	0.20	0.27	0.30	0.19
bmi	0.19	1.00	0.40	0.25	0.26	-0.37	0.41	0.45	0.39	0.59
bp	0.34	0.40	1.00	0.24	0.19	-0.18	0.26	0.39	0.39	0.44
s1	0.26	0.25	0.24	1.00	0.90	0.05	0.54	0.52	0.33	0.21
s2	0.22	0.26	0.19	0.90	1.00	-0.20	0.66	0.32	0.29	0.17
s3	-0.08	-0.37	-0.18	0.05	-0.20	1.00	-0.74	-0.40	-0.27	-0.39
s4	0.20	0.41	0.26	0.54	0.66	-0.74	1.00	0.62	0.42	0.43
s5	0.27	0.45	0.39	0.52	0.32	-0.40	0.62	1.00	0.46	0.57
s6	0.30	0.39	0.39	0.33	0.29	-0.27	0.42	0.46	1.00	0.38
progression	0.19	0.59	0.44	0.21	0.17	-0.39	0.43	0.57	0.38	1.00

```r
progression_correlations <- sort(
  cor_matrix[, "progression"],
  decreasing = TRUE
)
print(progression_correlations)
```

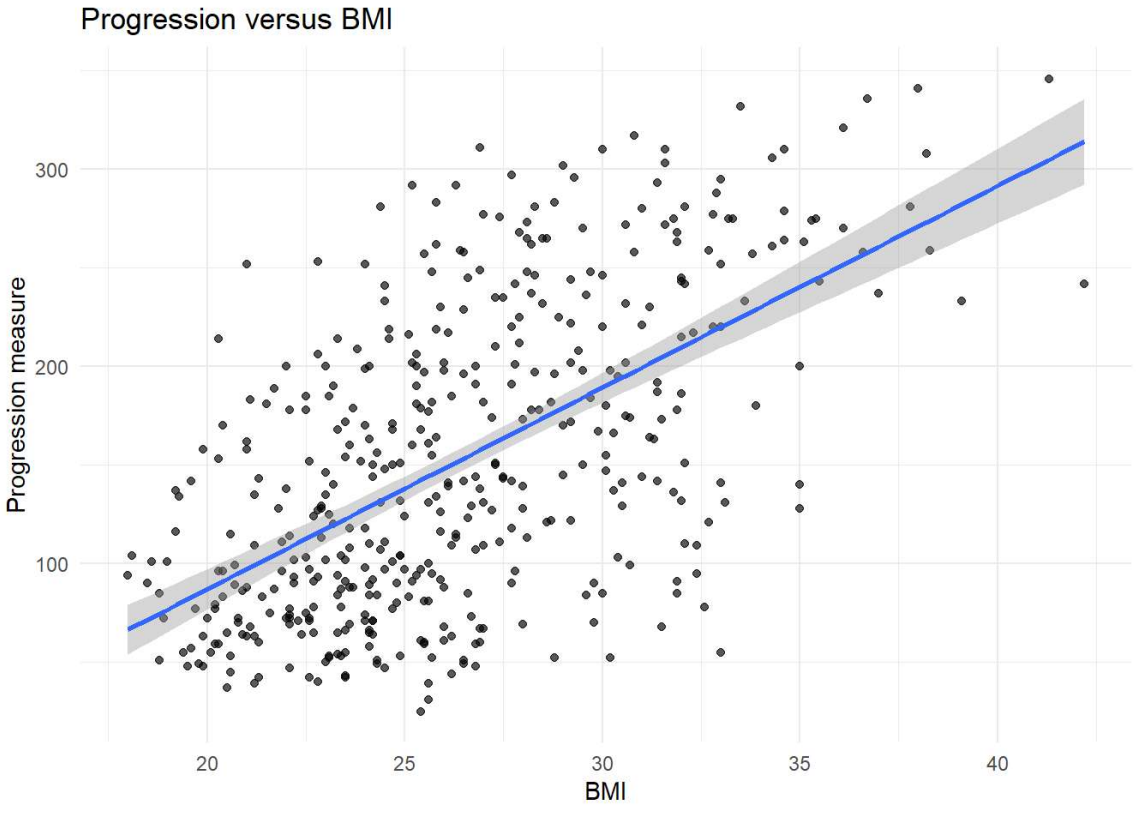
```
progression         bmi          s5          bp          s4          s6
     1.0000      0.5865      0.5659      0.4415      0.4305      0.3825
          s1         age          s2          s3
     0.2120      0.1879      0.1741     -0.3948
```

```r
heatmap(
  cor_matrix,
  Rowv = NA,
  Colv = NA,
  scale = "none",
  symm = TRUE,
  margins = c(7, 7),
  main = "Correlation heatmap"
)
```



```r
# Scatterplot 1: BMI and progression.
ggplot(diabetes, aes(x = bmi, y = progression)) +
  geom_point(alpha = 0.65) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Progression versus BMI",
    x = "BMI",
    y = "Progression measure"
  ) +
  theme_minimal()
```

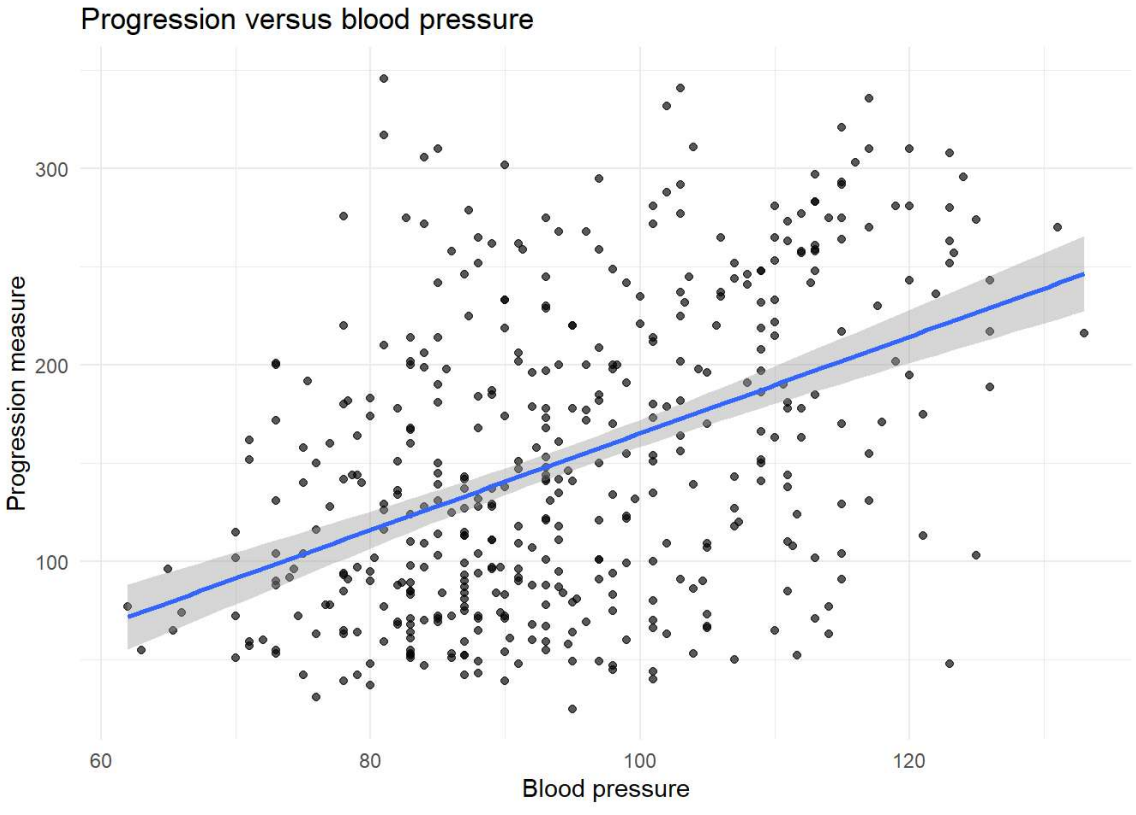
```
`geom_smooth()` using formula = 'y ~ x'
```




```r
# Scatterplot 2: blood pressure and progression.
ggplot(diabetes, aes(x = bmi, y = progression)) +
  geom_point(alpha = 0.65) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Progression versus blood pressure",
    x = "Blood pressure",
    y = "Progression measure"
```

```
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```
# ============================================================================
#  REGRESSION MODELS
# ============================================================================

# ---- Simple linear regression ------------------------------------------

simple_model <- lm(progression ~ bmi, data = diabetes)

# Standard model summary.
summary(simple_model)
```

```
Call:
lm(formula = progression ~ bmi, data = diabetes)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-164.92  -43.57   -8.65   46.34  154.88

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -117.773     18.019   -6.54  1.8e-10 ***
bmi           10.233      0.674   15.19  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 62.5 on 440 degrees of freedom
Multiple R-squared:  0.344, Adjusted R-squared:  0.342
F-statistic:  231 on 1 and 440 DF,  p-value: <2e-16
```

```r
# Tidy coefficient table, including confidence intervals.
tidy(simple_model, conf.int = TRUE)
```

```
# A tibble: 2 × 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)   -118.      18.0       -6.54 1.75e-10  -153.       -82.4
2 bmi             10.2      0.674     15.2  3.47e-42     8.91      11.6
```

```r
# Model-level statistics such as R-squared and adjusted R-squared.
glance(simple_model)
```

```
# A tibble: 1 × 12
  r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
      <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
1     0.344         0.342  62.5      231. 3.47e-42     1 -2454. 4914. 4926.
# ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
# Coefficients for writing the fitted equation.
coef(simple_model)
```

```
(Intercept)         bmi
    -117.77       10.23
```

```r
# ---- Multiple regression ------------------------------------------------

multiple_model <- lm(
  progression ~ bmi + bp + s5,
  data = diabetes
)
```

```r
summary(multiple_model)
```

```
Call:
lm(formula = progression ~ bmi + bp + s5, data = diabetes)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -334.881     25.878  -12.94  < 2e-16 ***
bmi            6.500      0.697    9.32  < 2e-16 ***
bp             0.903      0.217    4.17  3.7e-05 ***
s5            49.577      5.890    8.42  5.6e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 55.8 on 438 degrees of freedom
Multiple R-squared:  0.48,  Adjusted R-squared:  0.477
F-statistic:  135 on 3 and 438 DF,  p-value: <2e-16
```

```r
tidy(multiple_model, conf.int = TRUE)
```

```
# A tibble: 4 × 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept) -335.       25.9      -12.9  1.16e-32 -386.      -284.
2 bmi            6.50      0.697      9.32 5.55e-19    5.13       7.87
3 bp             0.903     0.217      4.17 3.74e- 5    0.477      1.33
4 s5            49.6       5.89       8.42 5.56e-16   38.0       61.2
```

```r
glance(multiple_model)
```

```
# A tibble: 1 × 12
  r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
      <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
1     0.480         0.477  55.8      135. 7.16e-62     3 -2403. 4815. 4836.
# ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
model_comparison <- bind_rows(
  glance(simple_model) %>% mutate(model = "Simple: BMI"),
  glance(multiple_model) %>% mutate(model = "Multiple: BMI + BP + S5")
) %>%
  select(model, r.squared, adj.r.squared, sigma)

print(model_comparison)
```

```
# A tibble: 2 × 4
  model                   r.squared adj.r.squared sigma
  <chr>                       <dbl>         <dbl> <dbl>
1 Simple: BMI                 0.344         0.342  62.5
2 Multiple: BMI + BP + S5     0.480         0.477  55.8
```

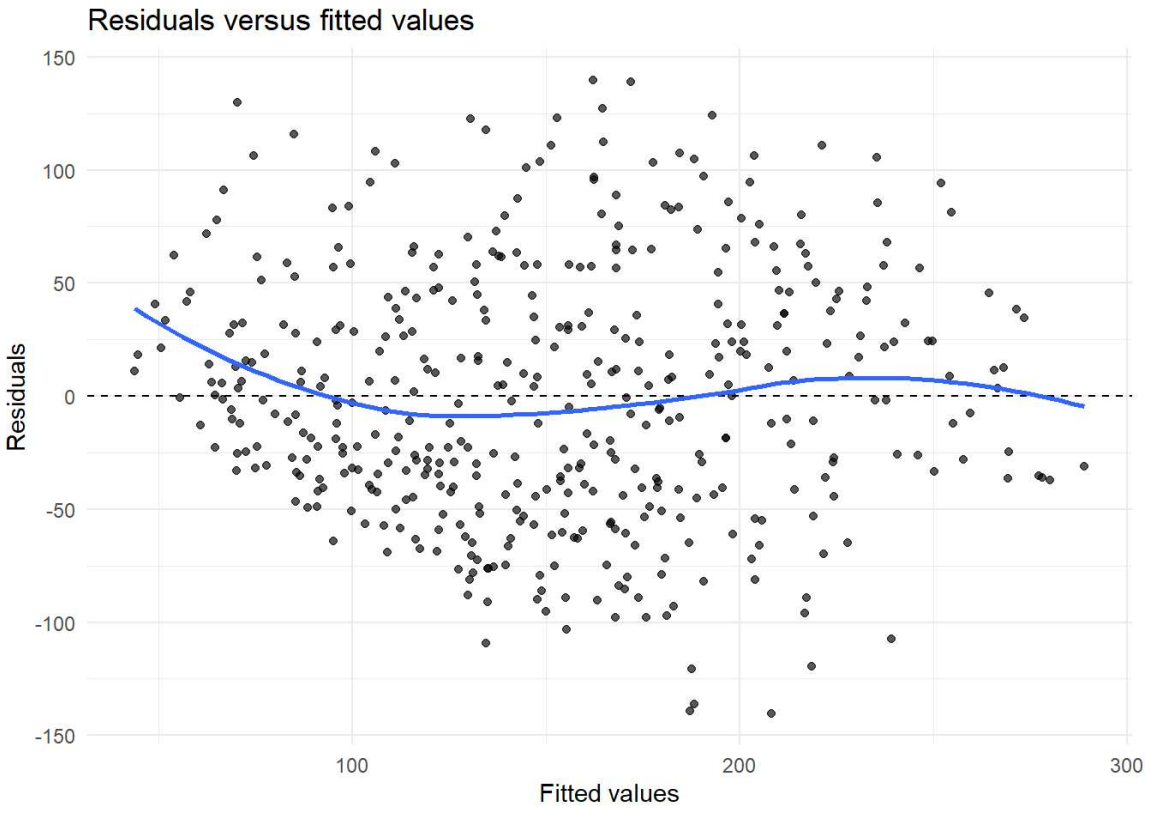
```r
# ---- Residuals versus fitted values --------------------------------------

diagnostic_data <- augment(multiple_model)

ggplot(diagnostic_data, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.65) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_smooth(se = FALSE) +
  labs(
    title = "Residuals versus fitted values",
    x = "Fitted values",
    y = "Residuals"
  ) +
  theme_minimal()
```



```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```r
# ---- Multicollinearity using VIF ----------------------------------------

# VIF can be calculated without installing another package.
manual_vif <- function(model) {
  X <- model.matrix(model)[, -1, drop = FALSE]  # remove intercept

  vif_values <- sapply(seq_len(ncol(X)), function(i) {
    target <- X[, i]
    others <- X[, -i, drop = FALSE]

    if (ncol(others) == 0) {
      return(1)
    }

    r2 <- summary(lm(target ~ others))$r.squared
    1 / (1 - r2)
  })

  setNames(vif_values, colnames(X))
}

manual_vif(multiple_model)
```

```
  bmi    bp    s5
1.345 1.274 1.342
```

```
# ---- Statistical uncertainty -------------------------------------------

coef_table <- tidy(multiple_model, conf.int = TRUE)
print(coef_table)
```

```
# A tibble: 4 × 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept) -335.       25.9      -12.9  1.16e-32 -386.      -284.
2 bmi            6.50      0.697      9.32 5.55e-19    5.13       7.87
3 bp             0.903     0.217      4.17 3.74e- 5    0.477      1.33
4 s5            49.6       5.89       8.42 5.56e-16   38.0       61.2
```

```
confint(multiple_model, level = 0.95)
```

```
                2.5 %   97.5 %
(Intercept) -385.7426 -284.020
bmi            5.1300    7.870
bp             0.4769    1.329
s5            38.0002   61.154
```